

Identity-Based Authentication Technique for IoT Devices

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Abstract—The Internet of Things (IoT) is a broad term used to describe many types of linked equipment that are both intelligent and interconnected through a network. These Internet of Things devices have become an essential part of our everyday lives. These devices, however, are still in the development stage. Because of their low computing capacity, they are susceptible to various cyberattacks like distributed denial of service (DDoS) attacks and denial of service (DoS) attacks. They disrupt the normal functioning of IoT devices and expose sensitive data; therefore, identifying and mitigating cyber-attacks on IoT devices is a critical area of study. In this regard, we proposed an IoT device-specific unique identity-based authentication method in this paper. Our proposed approach employs lightweight authentication procedures that use the unique IDs of IoT devices to provide identity-based encryption.

Index Terms—IoT, DDoS, DoS, Authentication

I. INTRODUCTION

The Internet of Things (IoT) is a network in which intelligent objects exchange real-time data (Figure 1). The Internet of Things is being used in a variety of areas, including energy management, intelligent transportation, smart grid management, and healthcare. Because of the broad range of applications, it is predicted that the sales revenues of Internet of Things devices would exceed \$14.4 trillion by 2022, and that there will be about 22 billion IoT devices by 2025 [1]. Because of their low power consumption and limited computing capacity, Internet of Things devices become susceptible to a variety of cyber attacks [2]–[5].

DDoS attacks are the deadliest and most often used cyber attacks, and we will discuss them in this article. It has

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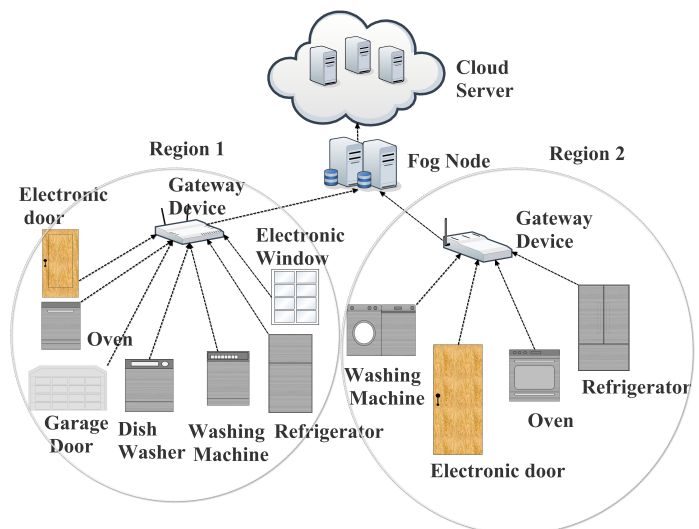


Fig. 1: IoT Environment

been almost a decade since the DDoS assault was invented, yet it is still very successful and is able to knock down the victim's server by depleting its resources. Most DDoS attacks are carried out by hackers who hack into Internet of Things (IoT) devices and use them to produce DDoS attack traffic. The Maria malware attack [6] is a current example of this kind of attack, in which the attacker generates 1.2 terabytes per second of attack traffic via IoT bots. Therefore, in recent years, researchers have proposed a variety of solutions to protect smart devices from distributed denial of service

(DDoS) attacks [7].

We developed a DDoS attack detection method in order to enhance the security of IoT devices. The following are the primary contributions to this essay. This article discusses how to use identity-based encryption (IBE) to build a secure data sharing mechanism and a cyber-attack detection system. The suggested method leverages the idea of identity-based encryption to manage access control to IoT devices, guaranteeing that no personally identifiable information is exposed.

Rest of the paper is organized as follows: Section II lists the latest research findings; Section III describes the proposed approach. Simulation results are represented in Section IV, and finally, Section V concludes the paper.

II. RELATED WORK

Recently, researchers proposed different techniques for authentication and security of IoT devices [8]–[14].

Authors in [15] pioneered the use of RSA cryptography in a two-way authentication security method for IoT devices. However, their method is limited to the most commonly used protocol, Datagram Transport Layer Security, and is well-suited for the Internet of Things platform. In [16], the authors discuss a privacy-preserving sharing system for IoT that is executed in the cloud. An IoT user may use the proposed method to encrypt the data and send it to the destination using identity-based encryption. In addition, IoT users have the ability to set specific access policies and create fine-grained access credentials that allow for delegating all encrypted data to the cloud to transform encrypted data meeting the access policies into new ciphertexts that may be read by an unknown receiver. This way, users of the Internet of Things may exchange data that has been outsourced to the cloud in a flexible and privacy-preserving manner. However, this scheme is secure against semitrusted clouds. Authors in [17] developed a new proximity-based authentication method for IoT devices. This plan asks smartphone users to hold their devices in one hand while using a predefined motion in front of them. Meanwhile, it keeps the user's distance from the gadget to a minimum. The only limitation of this approach is that it depends upon the motion sensors. Authors in [18] presented an anonymous and secure architecture for authentication that is more efficient and unlinkable. Their method needs frequent session key updates, which consumes a significant amount of time. Authors in [19] proposed a quick and error-free authentication and key agreement system for privileged devices that used channel state information as the standard secret key. Similarly, the distance has an effect on your strategy. A Function-based Access Control (FAC) strategy for IoT, in which an Identity-based Encryption (IBE) system is used, is proposed by the author in [20]. This suggested approach provides access control in an extremely granular manner, blocking illegal functions from running. The security analysis shows that the suggested system is secure, and the experimental findings indicate that the proposed method is effective against a variety of attack types.

III. PROPOSED METHODOLOGY

In this section, we give the details of our proposed approach. The identity-based protocol is used to authenticate Internet of Things devices in our proposed approach. Because each Internet of Things device has a unique ID value, it is feasible to create an identity-based communication architecture. Because of the suggested authentication protocol, Internet of Things devices are not impacted by various cyber-attacks such as distributed denial of service (DDoS). The following are the stages that make up the suggested identity-based authentication method.

A. Set-up Phase

The public parameters and master secret key are created in this step, and the plain text communication is ultimately encrypted into the ciphertext.

Algorithm 1: First Phase

Input: Input parameter (Sec)

Output: Encrypted Text

Begin:

$Sec \in \mathbb{Z}^*$

$\{para, Mas_s\} \xleftarrow{\$} \mathcal{Q}1^s$

$C_{text1} \xleftarrow{\$} k^{xU}$, where $U \leftarrow \{0, 1\}^*$

$C_{text2} \xleftarrow{\$} F^x$, $x \in \mathbb{Q}$,

$C_{text3} \xleftarrow{\$} \hat{v}(f, f)^x \cdot Meg_{text}$, where

$Meg_{text} \in \mathcal{M}$, message space

$C_{text} = (C_{text1}, C_{text2}, C_{text3})$

$C_{text} = (C_{text1}, C_{text2}, C_{text3})$, where $C_{text} \in \mathcal{C}$,

Cipher Space

End

- **Key Generation:** Generation of the public parameters and the master key occurs at this step. To keep our approach lightweight, we choose to use frameworks and lightweight technologies [20].

$$(para \in Public_r, Mas_s) \xleftarrow{\$} \mathcal{Q}(1)^S \quad (1)$$

$$para = (f, F) \quad (2)$$

$$Mas_s = y \quad (3)$$

where $F = f^x$; $x \in \mathbb{Z}_p^*$

- **Encryption:** With the assistance of the user's unique ID and public parameters, the plain text is encrypted.

$$C_{text1} = g^{xU} \quad (4)$$

$$C_{text2} = F^x \quad (5)$$

$$C_{text3} = \hat{e}(g, g)^x \cdot M_{text} \quad (6)$$

$$C_{text} = (C_{text1}, C_{text2}, C_{text3}) \quad (7)$$

where $r \in \mathbb{Z}_p^*$, $U \leftarrow \{0, 1\}^*$

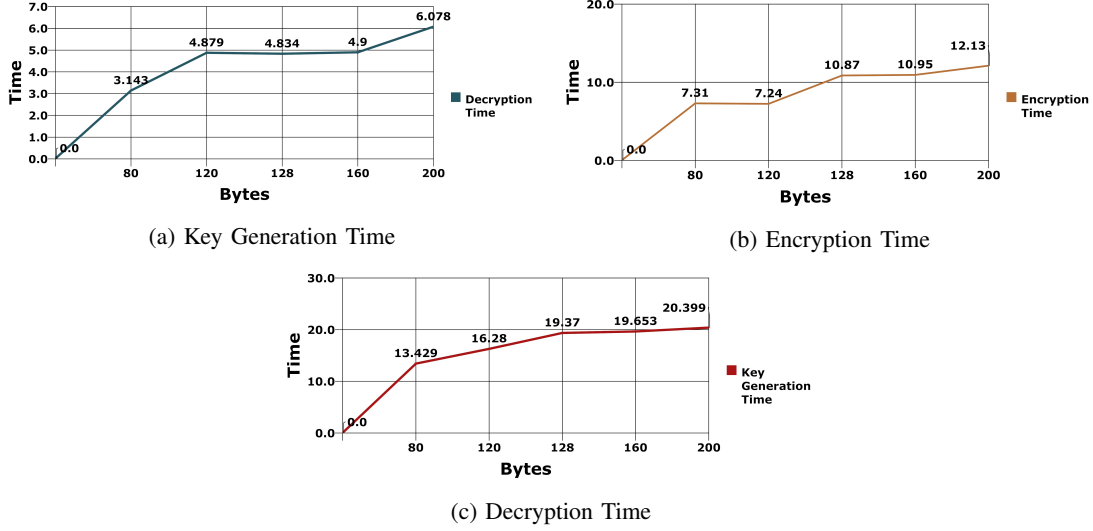


Fig. 2: Performance Evaluation of Proposed Approach

B. Private key Generation phase

The user's private key is produced in this step using the master key argument and the user's unique ID.

$$Per_k \leftarrow f^{\frac{1}{U+xt}} \quad (8)$$

where $t \in \mathbb{Z}_p^*$

C. Decryption Phase

Decryption is done in this phase plane (Equation 9).

$$M_{text} \in \mathcal{M} \leftarrow (Ct_1 Ct_2^t, P_{key}) \quad (9)$$

IV. RESULTS AND DISCUSSION

The technical and security underpinnings of our proposed IDB-S system are presented in earlier sections. This section discusses the implementation results of our IDB-S method. We constructed signature methods based on identity using a standard pairing library (PBC), on an Ubuntu 15.04 PC with 8 GB of RAM.

- We utilised five distinct test scenarios to assess the efficacy of our proposed technique in a variety of situations, calculating the key setup time, encryption time, decryption time, and total time for each scenario. The results for different test scenarios are shown in Figure 1.
- Key Set up stage:** This is the first stage of our proposed scheme. At this step, Algorithm 1 is used to create public parameters and master secret keys. In different test scenarios, Figure 2 displays the time required to choose public parameters and the master secret key. With the increase in the size of the selected key, there is a predictable trend for the key setup time to increase. As a result, generating a longer key takes longer. As a result, we must choose an optimum key length to ensure that little time is spent during the key creation process.
- Signature Time:** The second step of our suggested method is the signature stage; at this point, the service

TABLE I: Result Comparison

Approach	DDoS detection	Fine gained Access control	3 rd Party support
[21]	✗	✓	✓
[22]	✗	✗	-
Our Approach	✓	✓	✗

provider signs the data using its private key, which is kept hidden from the public. Various test scenarios involving different key lengths are roughly shown in Figure 2. The time needed to sign a secret message is proportional to the number of bits in the encryption key, as shown in the figure.

- Decryption Time:** This is the last step of our suggested strategy. The data receiver validates the digital signatures of the sent data using the service provider's public key. While the Figure 2 shows the amount of time required to perform the decryption procedure,

Table I compare our proposed approach with the other present approaches. From the Table I it is clear that our proposed approach works without the support of any 3rd party support and detect DDoS attacks.

V. CONCLUSION

The fact that IoT devices handle users' personal and private information encourages hackers to attempt to hack them using a variety of cyber-attacks, and the fact that IoT devices have limited power and memory makes it difficult to defend against these cyber-attacks. In order to prevent access with excessive privileges in the Internet of Things, we offer the identity-based Access Control system. Since the connection among IoT devices is verified using our suggested method, these devices may avoid being subjected to various kinds of flooding attacks such as DDoS.

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